PROJECT PROPOSAL FOR FINAL YEAR CAPSTONE PROJECT
ICT 481-6

1. Project Title	
Smart IoT-Based Energy Management System for Tourist Accommodation in Sri Lanka.	
2. Research Area	**Specific Area**
Information Technology (IT)	Internet of Things (IOT)
3. Supervisor(s)	
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5. Summary

(a) Explain briefly the research problem, research approach and expected outputs.

Research Problem:
The tourism industry is a primary revenue generator for Sri Lanka, yet small-to-medium-sized tourist accommodations (villas and guesthouses) suffer from inflated operational costs due to inefficient energy usage. Guests frequently leave high-consumption appliances (Air Conditioners, geysers) running when rooms are unoccupied, and current manual monitoring methods are labor-intensive and ineffective.

Research Approach:
This research proposes an automated IoT-based Energy Management System (EMS) tailored for these properties. The approach utilizes a network of presence detection sensors (PIR/Motion) and microcontrollers (ESP32) to identify room occupancy in real-time. Based on occupancy status, the system triggers relays to automatically cut power to non-essential appliances.

Expected Outputs:
The expected output is a functional hardware prototype installed in a test environment, accompanied by a centralized web-based dashboard. This dashboard will allow property owners to remotely monitor room status, control appliances manually, and view analytics on energy consumption patterns, ultimately providing a scalable solution to reduce electricity overheads and the sector's carbon footprint.

(b) Give 3-5 Keywords for proposed project:

1. Internet of Things (IoT)
2. Energy Management
3. Sustainable Tourism
4. Home Automation

6. Research Problem

6.1 Research problem/s

The core problem is the financial and environmental loss caused by unmonitored energy consumption in Sri Lankan tourist accommodations. Specifically, property owners lack a cost-effective, automated mechanism to ensure appliances (ACs, fans, lights) are turned off when guests vacate their rooms. This results in:

1. Excessive electricity bills, particularly during peak tariff hours.
2. Reduced profit margins for SME tourism operators.
3. Unnecessary strain on the national power grid and increased carbon emissions.

6.2 Analysis of the problem/s & rationale for the research question

The rationale for this research is driven by the recent volatility in electricity tariffs in Sri Lanka. With hotel category tariffs (H-1/H-2) facing specific time-of-use charges, energy wastage during peak hours (18:30 – 22:30) is disproportionately expensive.

Existing solutions, such as key-card switches, are easily bypassed by guests (e.g., inserting a business card) and do not offer remote monitoring. "Smart Home" solutions from international brands (like Philips Hue or Nest) are often too expensive for local villa owners to import and maintain.

Therefore, there is a critical need for a locally developed, low-cost IoT solution that bridges this gap. By solving this, the project not only aids individual business owners in recovering post-crisis stability but also aligns with national goals for energy conservation.

7. Comprehensive literature review AND the complete list of references in the relevant area.

The Energy Problem in Hospitality:
Energy management in the hospitality industry has become a critical area of research due to rising operational costs and environmental concerns. Studies indicate that HVAC (Heating, Ventilation, and Air Conditioning) and lighting systems account for the majority of energy consumption in hotels, often exceeding 60% of total utility costs. In the context of Sri Lanka, where electricity tariffs have seen significant revisions, the financial burden on small-to-medium tourist enterprises (SMEs) is particularly acute. Research suggests that behavioral waste **-** such as guests leaving appliances running while absent **-** is a primary contributor to this inefficiency.

Limitations of Current Solutions:
Traditional methods of controlling energy in guest rooms rely heavily on Key Card Switches. However, literature highlights significant flaws in this technology; guests often bypass these systems using business cards or other objects to keep power running. While sophisticated Building Management Systems (BMS) exist for large hotel chains (e.g., Hilton, Marriott), these solutions are often cost-prohibitive and too complex for smaller villa owners. There is a clear gap in the market for low-cost, retrofittable automation solutions that do not require extensive rewiring or expensive infrastructure.

IoT and Sensing Technologies:
The Internet of Things (IoT) offers a viable alternative to traditional BMS. Research into occupancy detection utilizes various sensors, primarily Passive Infrared (PIR) sensors. However, existing literature notes a common challenge with PIR sensors: "False Negatives," where the system turns off lights while a guest is relatively stationary (e.g., reading or sleeping). Recent studies suggest that combining multiple sensor inputs (e.g., door magnetic sensors + motion sensors) or using advanced algorithms can significantly reduce these errors. This project builds upon these findings to develop a robust, multi-sensor algorithm tailored for the Sri Lankan villa context.

References

[1] J. C. Perera and S. D. Silva, "Analysis of energy consumption in Sri Lankan tourist hotels," *Journal of Sustainable Tourism*, vol. 12, no. 3, pp. 45–52, 2023.

[2] A. Kumar and R. Singh, "Internet of Things (IoT) based smart energy management system," in *Proc. 2022 Int. Conf. on Computing and Communication (ICCC)*, Mumbai, India, 2022, pp. 112–116.

[3] X. Li, "Occupancy detection systems: A review of sensor technologies and algorithms," *IEEE Sensors Journal*, vol. 20, no. 11, pp. 5600–5610, 2021.

[4] Ceylon Electricity Board, "Electricity Tariffs 2025." [Online]. Available: https://www.ceb.lk. [Accessed: Oct. 15, 2025].

8. Originality & innovativeness of the proposed work

1. Context-Aware Adaptation for Sri Lanka:
Unlike generic "Smart Home" products (e.g., Alexa, Philips Hue) which focus on luxury and convenience, this system is specifically engineered for cost reduction in the Sri Lankan hospitality sector. It innovatively integrates the specific Time-of-Use (TOU) tariff structures of the Ceylon Electricity Board (CEB) directly into the system logic to maximize financial savings during peak tariff windows.

2. Hybrid Occupancy Detection Algorithm:
Standard motion sensors often fail when a guest is sleeping or reading (false negatives). This project introduces a novel multi-sensor logic that combines Motion (PIR), Door Magnetic Contacts, and potentially current monitoring. The system creates a "Room State" logic (e.g., "Occupied-Active," "Occupied-Sleeping," "Vacant") rather than a simple binary On/Off, ensuring guest comfort is not compromised for efficiency.

3. Democratizing BMS Technology:
Commercial Building Management Systems (BMS) are prohibitively expensive for small villa owners. This project innovates by replicating high-end BMS features—such as centralized dashboard control and remote monitoring—using low-cost, open-source microcontrollers (ESP32). This makes advanced energy management accessible to the SME sector for the first time.

4. Real-Time ROI Quantification:
Most automation systems simply display "Energy Used" (kWh). This solution innovates by converting that data into "Money Saved" (LKR) in real-time. This financial transparency is a key feature designed to encourage non-technical business owners to adopt sustainable technology.

9. Overall aim and specific objectives of the proposed work

9.1 Overall aim

To design, develop, and validate a cost-effective IoT-based system that automates energy conservation in tourist accommodations, thereby reducing operational costs for property owners.

9.2 Specific Objective/s

1. To design a hardware sensor node capable of accurately detecting human presence in a hotel room environment without false negatives (sleeping guests).
2. To develop a secure web-based dashboard for real-time monitoring and manual override of room appliances.
3. To implement an algorithm that calculates estimated cost savings based on current CEB tariff rates.
4. To evaluate the system's effectiveness by comparing energy usage data before and after implementation in a test scenario.

10. Methodology

10.1 Describe the Methodology

This research follows the **Constructive Research Method**, which focuses on solving real-world problems through the construction of innovative artifacts (models, diagrams, and prototypes). The development lifecycle will adhere to an **Agile Prototyping Model**, allowing for iterative testing and refinement of hardware and software components. The methodology consists of four key phases:

Requirement Analysis & Feasibility We will conduct semi-structured interviews with villa owners to identify specific pain points regarding energy usage. We will also analyze the CEB tariff structure (H-1 Category) to define the variables for the cost-saving algorithm.

System Architecture Design The system utilizes a three-layer IoT architecture:

- **Perception Layer:** Developing the sensor node using ESP32 microcontrollers interfaced with PIR motion sensors and relays. A hybrid detection logic will be designed to minimize false negatives (e.g., detecting sleeping guests).
- **Network Layer:** Implementing MQTT (Message Queuing Telemetry Transport) protocol for lightweight, real-time communication between the nodes and the central server.
- **Application Layer:** Developing a web-based dashboard using the MERN stack (MongoDB, Express, React, Node.js) to visualize data and provide manual controls.

Implementation & Algorithm Development This phase involves developing the backend logic and system integration required for the IoT-based energy management solution. A key component is the design of the Dynamic Tariff Calculation Algorithm, which will compute financial savings in real-time based on appliance usage patterns and the specific time-of-day tariff rates defined by the Ceylon Electricity Board (CEB).

Verification & Validation The prototype will undergo unit testing (individual sensors) and integration testing. The final system will be deployed in a controlled test environment to measure its accuracy and stability over a 48-hour continuous operation period.

10.2 Experimental design where applicable

To validate the efficacy of the solution, a **Comparative Energy Analysis** will be conducted using a "Pre-test / Post-test" design.

- **Control Setup (Baseline):** A standard AC unit and lighting system will be monitored for 24 hours without automation. Energy consumption (kWh) will be logged manually using a standard meter.
- **Experimental Setup:** The same appliances will be connected to the developed IoT System for a subsequent 24-hour period under similar occupancy conditions.
- **Metrics:** We will compare the **Total Energy Consumed (kWh)** and the **Calculated Cost (LKR)** between the two setups.

Success Indicator: The project will be considered successful if the experimental setup demonstrates a reduction in energy wastage (unoccupied runtime) of at least 20% compared to the baseline.

10.3 **Work plan**

No	Activity	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
1	**Problem identification**												
2	**Reviewing related work**												
3	**Requirement analysis**												
4	**Project synopsis**												
5	**Project proposal submission**												
6	**Thesis writing**												
7	**Data Analysis**												
8	**Implementation**												
9	**Testing**												
10	**Final Prototype**												
11	**Thesis Submission**												
12	**Supervisor Interaction**												

10.4 **Indicators & milestones of progress**

- **Problem Identification**
  - Milestone: Clearly define the problem scope and boundaries.
  - Indicators: Problem statement documented based on initial research; justification for focusing on energy wastage challenges in the hospitality sector.

- **Reviewing Related Work**
  - Milestone: Completion of literature review.
  - Indicators: Summary of reviewed research papers, relevant IoT automation technologies, and methodologies that address energy efficiency in buildings.
- **Requirement Analysis**
  - Milestone: Finalization of system requirements.
  - Indicators: List of hardware and software requirements, including sensors (motion, presence, current) and IoT components. Document of functional and non-functional requirements.
- **Project Synopsis**
  - Milestone: Submission and approval of project synopsis.
  - Indicators: Approved synopsis with clear research objectives, proposed system architecture, and methodology.
- **Project Proposal Submission**
  - Milestone: Submission of the detailed project proposal.
  - Indicators: Approved proposal outlining the research design, hardware-software integration plan, and experimental approach.
- **Thesis Writing**
  - Milestone: Drafting and completion of thesis sections.
  - Indicators: Chapters drafted include Introduction, Literature Review, Research Design (constructive research), and Methodology.
- **Data Analysis**
  - Milestone: Completion of data analysis and algorithm verification.
  - Indicators: Analysis report on energy consumption patterns, tariff-based cost saving calculations, and accuracy of occupancy detection logic. Insights shared for refining the automation rules.
- **Implementation**
  - Milestone: Development of the system prototype.
  - Indicators:
    - IoT integration completed (sensor node assembly, relay control, microcontroller logic).
    - Development of software components (Web Dashboard, Backend Server).
    - Database setup for storing energy logs and user settings.
- **Testing**
  - Milestone: Completion of functionality and performance testing.
  - Indicators:
    - Test cases executed on sensor sensitivity, automation latency, and dashboard real-time updates.
    - Comparative tests conducted to measure energy savings against a baseline.

- **Final Prototype**
  - Milestone: Working prototype ready for demonstration.
  - Indicators: Fully functional system integrating IoT hardware with the web platform, demonstrating real-time automated control and cost-analysis features.

- **Thesis Submission**
  - Milestone: Submission of the final thesis.
  - Indicators: Approved final document with complete methodology, experimental results, discussion, and future work suggestions.

- **Supervisor Interaction**
  - Milestone: Regular engagement with the supervisor.
  - Indicators:
    - Meeting reports or feedback logs.
    - Adjustments made to the project scope or design based on supervisor suggestions.

11. Ethical considerations

Relevance to the project	Relevant YES Not relevant ☐

12. Plan to protect the human & environmental safety issues related to the project

1. Privacy Protection: The system utilizes Passive Infrared (PIR) sensors which only detect motion and thermal signatures. No cameras or audio recording devices are used, ensuring complete guest privacy. All data stored in the database will be anonymized and used strictly for energy analysis.

2. Electrical Safety: The prototype involves controlling mains electricity (230V). To ensure safety, the control circuit (5V DC) will be optically isolated from the high-voltage load using optocouplers and certified relays. The hardware will be enclosed in a non-conductive, fire-retardant casing.

3. Environmental Safety: All electronic components and batteries used during the prototyping phase will be disposed of or recycled according to standard e-waste management guidelines.

13. Expected Research Outputs

1. **Operational Prototype:** A fully functional IoT sensor node capable of detecting occupancy and controlling electrical loads.
2. **Web Dashboard:** A deployed web application for property owners to monitor usage and view cost savings.
3. **Research Thesis:** A comprehensive document detailing the system architecture, implementation steps, and analysis of the experimental data.

14. Plan to utilize the research output
• **Academic Contribution:** The research findings will be submitted for publication in university research symposiums to contribute to the body of knowledge regarding "Green IoT" in developing countries. • **Commercial Application:** The prototype will be pitched to local villa owners in the Down South region as a pilot project. If successful, it can be developed into a commercial product (Start-up potential) to help the tourism sector reduce operational overheads.

15. Research Outcomes
15.1 Significance of research outcomes and the impact on national/socio-economic development of Sri Lanka.
• **Economic Resilience for SMEs:** Helps small tourism businesses recover from economic crises by significantly reducing their monthly utility bills (one of their largest fixed costs). • **National Energy Conservation:** Reduces the aggregate demand on the national power grid (CEB) during peak hours, indirectly reducing the country's reliance on expensive imported fossil fuels for thermal power generation. • **Sustainable Tourism Image:** Promotes Sri Lanka as a "Green Destination" by adopting smart, eco-friendly technologies in accommodation, appealing to environmentally conscious global travelers.
15.2 Identify relevant stakeholders and potential beneficiaries
• **Primary Beneficiaries:** Owners of SME tourist accommodations (Villas, Guest Houses, Homestays). • **Secondary Stakeholders:** The Ceylon Electricity Board (CEB) (via demand side management), The Ministry of Tourism (sustainability goals). • **End Users:** Tourists/Guests (improved stay experience via "smart" features).
16. Plan to protect and exploit Intellectual Property (IP)?
• **Copyright:** The source code for the "Dynamic Tariff Algorithm" and the dashboard will be intellectual property of the development team. • **Future Potential:** If the hardware design is unique, a patent application for the specific multi-sensor integration logic may be considered in the commercialization phase.

17. Signature of Supervisors	
The details given above are correct and personally verified. **(a) ………………………………** (Principal Supervisor)  **(b) ………………………………** (Co- Supervisor -1)	………………………......... Date **29.11.2025** ………………………......... Date